3. When a magnet is broken into pieces, each piece behaves like a magnet with poles at its ends.
4. Two poles of a magnet have pole strength equal to one another.
5. The length of the bar magnet is called geometrical length and the length between two magnetic poles in a bar magnet is called magnetic length. Magnetic length is always slightly smaller than geometrical length. The ratio of magnetic length and geometrical length is $\frac{5}{6}$.

$$\frac{Magnetic\ length}{Geometrical\ length} = \frac{5}{6} = 0.833$$

EXAMPLE 3.2

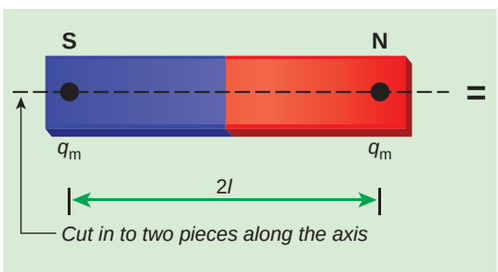
Let the magnetic moment of a bar magnet be $\vec{p}_m$ whose magnetic length is d = 2l and pole strength is q_m. Compute the magnetic moment of the bar magnet when it is cut into two pieces

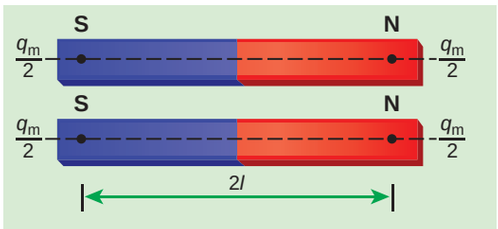
(a) along its length

(b) perpendicular to its length.

Solution

(a) a bar magnet cut into two pieces along its length:





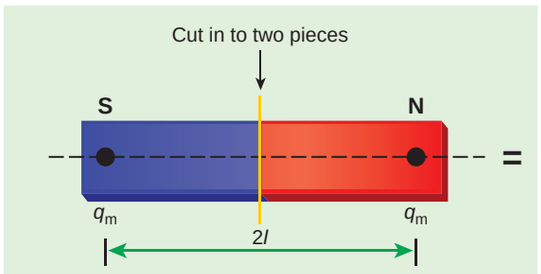
When the bar magnet is cut along the axis into two pieces, new magnetic pole strength is $q'_m = \frac{q_m}{2}$ but magnetic length does not change. So, the magnetic moment is

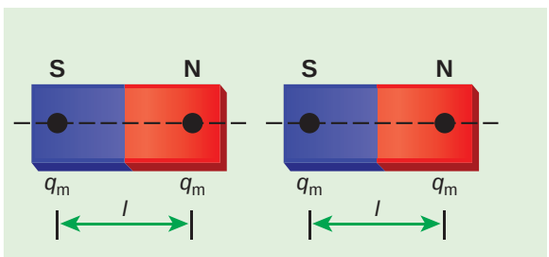
$$p'_m = q'_m 2l$$

$$p'_m = \frac{q_m}{2} 2l = \frac{1}{2}(q_m 2l) = \frac{1}{2} p_m$$

In vector notation, $\vec{p}'_m = \frac{1}{2}\vec{p}_m$

(b) a bar magnet cut into two pieces perpendicular to the axis:





When the bar magnet is cut perpendicular to the axis into two pieces, magnetic pole strength will not change but magnetic length will be halved. So the magnetic moment is

$$p'_m = q_m \times \frac{1}{2}(2l) = \frac{1}{2}(q_m \cdot 2l) = \frac{1}{2} p_m$$

In vector notation $\vec{p}'_m = \frac{1}{2}\vec{p}_m$

UNIT-3(XII-Physics_Vol-1).indd 135 04-03-2019 10:33:22